

Method

Network flow

The implementation follows four operations:

1. extract a Common feature F and a fine-grained detail feature from RGB;
2. form Part-structure information S and Visual evidence E ;
3. form and progressively correct class-specific symptom candidates; and
4. interpret the corrected candidates using Route assignment, Spatial Support, and Route-wise composition before final refinement.

The network is implemented in `models/network.py`. The paper configuration uses a RepViT-M1.1 encoder, a lightweight multiscale decoder with 96 output channels, and 64-channel prediction heads.

Notation and outputs

Symbol	Definition	Channels
F	Common feature	96
S	Fish Region, Part Map, Head-to-Tail Direction, and Zone Map	15
E	Redness, Shape, Lesion, and Unaffected Surface	4
R_0	Initial symptom candidate response	8
N	Signed-corrected candidate response	8
A	Gated residual correction term	8
$C = N + A$	Corrected symptom candidate response	8
Z	Routed semantic logits	8
Y	Auxiliary semantic mask	9 classes

R_0 , N , and C are independent class responses. Multiple symptom classes may therefore remain active at one pixel. Y is mutually exclusive over the background and eight symptom classes.

Part-structure information S

Fish Region combines the Common feature with an RGB-derived foreground prior. The predicted fish geometry defines a principal axis and two endpoint priors. An endpoint-role head fixes the head-to-tail sign, after which the Part Map is formed over background, body, fin, caudal fin, and mouth.

Zone Map is derived from the soft Part Map, fish boundary, body contact, and longitudinal position. Its channel order is:

1. body;
2. mouth;

3. fin tip;
4. fin middle;
5. fin base;
6. caudal-fin tip;
7. caudal-fin middle; and
8. caudal-fin base.

The maps remain continuous in the network. The part reference is fixed after the Part-structure formation stage and supplies the spatial input used by the later candidate and route stages.

Visual evidence E

Redness compares each pixel with the global fish color and a local color mean, then combines that prior with the Common feature, Fish Region, and Part Map. Shape uses common and fine-grained features together with fish-boundary and appendage-zone cues. Lesion uses common and fine-grained local contrast and variance inside the fish. Unaffected Surface combines both feature streams, Part-structure information, and the three positive Visual evidence responses.

Before candidate formation, Lesion is attenuated by detached Unaffected Surface:

$$E_L^{\text{eff}} = E_L \text{ clip}(1 - 0.75E_U, 0.05, 1)$$

Visual evidence is detached from the candidate and route paths in Stages 3-6. Its target construction is given in [SUPERVISION.pdf](#).

Initial symptom candidate response R_0

R_0 is predicted from the Common feature, fine-grained detail feature, Part Map, Zone Map, Redness, Shape, and effective Lesion. It contains one logit map for each symptom class.

Signed candidate correction

For class c and cue k , the predefined relationship is required support, optional support, inhibition, or unused. Cue activation is

$$a_{c,k}(x) = \sigma \left(\frac{q_k(x) - \tau_{c,k}}{t_{c,k}} \right), \quad t_{c,k} = \text{softplus}(\hat{t}_{c,k})$$

With nonnegative learned strengths, the implemented support and inhibition are

$$U_c^+(x) = \sum_{k \in \mathcal{K}_c^{\text{req}}} w_{c,k}^{\text{req}} \log a_{c,k}(x) + \sum_{k \in \mathcal{K}_c^{\text{opt}}} w_{c,k}^{\text{opt}} \log(1 + a_{c,k}(x)) + w_c^Q \log Q_c(x),$$

$$U_c^-(x) = \sum_{k \in \mathcal{K}_c^{\text{inh}}} w_{c,k}^{\text{inh}} a_{c,k}(x) + \sum_{j \in \mathcal{C}_c^{\text{comp}}} \kappa_{c,j} \text{softplus}(\tilde{N}_j(x) - \tilde{N}_c(x) + \mu_{c,j})$$

Here Q_c is the class-compatible Zone Map and the competing-class term is used within the fin and caudal-fin class groups. The signed response is

$$N_c(x) = R_{0,c}(x) + U_c^+(x) - U_c^-(x)$$

Symptom class	Required support	Optional support	Main inhibition
Body lesion	body, Lesion	Redness	non-body locations
Mouth ulcer	mouth, Lesion	Redness	non-mouth locations, Redness alone
Fin or caudal deformity	corresponding tip, Shape	Lesion	middle or base, Lesion alone
Fin or caudal necrosis	corresponding middle, Lesion	Redness, Shape	tip, Shape alone
Fin or caudal base necrosis	corresponding base, Lesion	Redness, Shape	tip, Shape alone

Gated residual correction

The Common feature, detail feature, Zone Map, $\sigma(R_0)$, $\sigma(N)$, Redness, Shape, and Lesion are fused to predict a gate logit G_c and signed residual Δ_c . This first produces

$$C_{g,c}(x) = N_c(x) + \sigma(G_c(x))\Delta_c(x)$$

In the FULL configuration, C_g is combined with the class-compatible Zone Map, part-conditioned response, Visual evidence, and learned candidate context to form C . The exposed correction term is $A = C - N$, so the public responses satisfy $C = N + A$.

Part-routed class interpretation

Route assignment is formed over background, body, fin, caudal fin, and mouth. The four foreground responses exposed by the public interface are ordered as body, mouth, fin, and caudal fin. The learned logits are added to a strong Part-Map log prior.

Spatial Support combines the Common feature, detail feature, Route assignment, Zone Map, C , and E . Its public order follows the eight Zone Map channels. Fin-base and caudal-fin-base support retain attachment to the body route.

Each route then predicts only its allowed classes:

Route	Allowed symptom classes
Body	body lesion
Mouth	mouth ulcer
Fin	fin deformity, fin necrosis, fin base necrosis
Caudal fin	caudal deformity, caudal necrosis, caudal base necrosis

Route-wise composition places the route responses into the common eight-class space. Corrected candidates are retained where route confidence is weak, giving the Routed semantic logits Z .

Semantic output and final refinement

Z , route-class support, Fish Region, and Unaffected Surface form native background-plus-symptom logits. Final refinement uses those logits with common and fine-grained features, Zone Map, fish boundary, and Visual evidence to predict foreground and class residuals. The argmax of the refined nine-class logits is the Auxiliary semantic mask Y .

```
outputs = model(images, return_aux=False)
R0 = outputs["R0"]
N = outputs["N"]
A = outputs["A"]
C = outputs["C"]
Z = outputs["Z"]
Y = outputs["Y"]
```

Use `return_aux=True` to obtain the named Part-structure, Visual evidence, Route assignment, Spatial Support, native semantic, and refinement tensors.